COVID-Era Changes in U.S. Human Trafficking

A Multi-method Quantitative Analysis

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Introduction

COVID-19 impacted nearly every aspect of life across the globe. Human traffickers were quick to adapt (Polaris Project, 2023). But how did they adapt? Specifically, did they change the type of exploitation, the means of control, or was there a shift in the relationship between victim and exploiter? This report utilizes advanced analytical methods to explore these questions.

Data and Methods

Data Source

- Dataset was downloaded from the Counter Trafficking Data Collaborative website in January 2026.

Dataset Description

- Single dataset containing 257,969 rows across 27 features.
- Data from international reports is included.
- YearOfRegistration spanning from 2002 to 2023.
- Pre-processing included:
    - Filtering CountryOfExploitation to the USA.
    - Addition of a pre-/post-COVID helper column.
        - Pre-COVID years 2002-2019
        - Post-COVID years 2020-2023
        - To maintain consistent comparisons, the number of rows in the years 2002-2015 were evaluated. With only 3,367 rows, it was determined they would not significantly skew the results and they were left in the dataset.
    - Converting null values to 0.
    - Converting datatypes.
    - After pre-processing, the dataset contained 116,448 rows across 27 features.
- For the machine learning model, pre-processing was performed again on a fresh copy of the dataset and included:
    - Filtering to the USA, then dropping the CountryOfExploitation feature to eliminate noise in the clusters.
    - Converting datatypes.
    - Dropping rows where YearOfRegistration was missing.
    - Assessing feature missingness and dropping features with high percentages of missing values.
    - Assessing missing values per row and dropping rows with four or more missing values.
    - Utilizing OneHotEncoding to handle categorical variables.
    - Addition of missing indicator columns so null values could be filled with 0.
    - Applying feature scaling prior to PCA transformation.
    - After pre-processing, the machine learning dataset contained 78,707 rows across 21 features (6 features prior to OneHotEncoding and missing indicator columns being added).

Hypothesis

- The null hypothesis (H_0) asserts there is no statistical significance in trafficking patterns before and after COVID-19.
- The alternative hypothesis (H_1) asserts statistical significance exists in trafficking patterns before and after COVID-19.

Analytical Methods

- Chi-square tests: Chi-square results were evaluated for p-values < .05 to evaluate statistical significance, degrees of freedom to equal 1 since the contingency tables are 2X2, and expected frequencies > 5 to understand if additional testing should be conducted.
- PCA: PCA was applied to reduce dimensionality and principal components were analyzed to identify high and low PC scores.
- K-Means: K-means clusters were evaluated by comparing principal components against a cluster centroid heatmap.

Analysis and Findings

Chi-square testing was conducted on three Types of Exploitation, eight Means of Control, and four Relationships to Exploiter.

Type of Exploitation

Type of Exploitation	Forced Labor	Sexual Exploitation	Other Exploitation
% Change Post COVID Raw	-23.48%	-29.85%	170.00%
% Change Post COVID Normalized	0.28%	-4.77%	0.04%
% Of Total Post COVID	19.28%	59.68%	0.06%
% of Total Pre COVID	19.08%	64.45%	0.02%
Statistic	0.64358	260.37338	12.27621
P-value	0.422415	1.4227E-58	0.00045
Statistically Significant?	No	Yes	Yes
Degrees of Freedom	1	1	1
EF Post COVID-False	38119.225	17733.17	47142.0525
EF Post COVID-True	9038.7747	29424.83	15.9474829
EF Pre COVID-False	50321.775	23409.83	62232.9475
EF Pre COVID-True	11932.225	38844.17	21.0525171

- Forced Labor changes were not statistically significant.
  - Fail to reject H_0.
  - With a p-value > .05 there is insufficient evidence to suggest COVID impacted Forced Labor.
- Sexual Exploitation changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted Sexual Exploitation.
  - A decrease of 4.77% in the normalized percent of change represents a moderate shift post-COVID.
- Other Exploitation changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted Other Exploitation.
  - An increase of .04% in the normalized percent of change represents a small shift post-COVID.

While changes in both Sexual Exploitation and Other Exploitation are statistically significant, the large reductions in the raw percentages and much smaller normalized percent change show the application to real world shifts may be limited. Of note, the pre-COVID True population for Other Exploitation was 10, while the post-COVID True population was 27, contributing to the exaggerated raw 170% increase. The small population in this feature would warrant further research, and may not be as practically significant as indicated by the raw percent of change pre- and post-COVID.

Means of Control

Means of Control	Debt Bondage	Threats	Abuse	False Promises	Drugs and Alcohol	Deny Basic Needs	Excess Work Hours	Withhold Documents
% Change Post COVID Raw	-35.10%	-28.26%	-30.51%	-38.54%	-40.28%	-42.54%	-20.76%	-24.80%
% Change Post COVID Normalized	-1.87%	-1.04%	-2.01%	-0.85%	-1.98%	-4.43%	0.21%	-0.03%
% Of Total Post COVID	11.16%	18.53%	22.30%	3.67%	7.37%	13.92%	4.78%	3.51%
% of Total Pre COVID	13.03%	19.57%	24.31%	4.52%	9.35%	18.35%	4.57%	3.54%
Statistic	87.02116	18.58735	60.16863	48.75932	134.85233	383.37989	2.63225	0.04415
P-value	1.0737E-20	1.62293E-05	8.7068E-15	2.89379E-12	3.5566E-31	2.286E-85	0.10471	0.83356
Statistically Significant?	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Degrees of Freedom	1	1	1	1	1	1	1	1
EF Post COVID-False	41391.9074	38140.77594	36104.2393	45200.33871	43152.16469	39403.644	44958.5404	45495.1516
EF Post COVID-True	5766.0926	9017.22405	11053.7607	1957.66128	4005.8353	7754.3558	2199.4596	1662.84835
EF Pre COVID-False	54642.0926	50350.22405	47661.7607	59669.66128	56965.8353	52017.356	59350.4596	60058.8484
EF Pre COVID-True	7611.90739	11903.77594	14592.2393	23584.33871	5288.16469	10236.644	2903.54039	2195.15164

- Debt Bondage changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted Debt Bondage.
  - A decrease of 1.87% in the normalized percent of change represents a small to moderate shift post-COVID.
- Threat changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted Threat.
  - A decrease of 1.04% in the normalized percent of change represents a small shift post-COVID.
- Abuse changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted Abuse.
  - A decrease of 2.01% in the normalized percent of change represents a moderate shift post-COVID.
- False Promises changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted False Promises.
  - A decrease of 0.85% in the normalized percent of change represents a small shift post-COVID.
- Drugs and Alcohol changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted Drugs and Alcohol.
  - A decrease of 1.98% in the normalized percent of change represents a small to moderate shift post-COVID.
- Deny Basic Needs changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted Deny Basic Needs.
  - A decrease of 4.43% in the normalized percent of change represents a moderate shift post-COVID.

- Excess Work Hours changes were not statistically significant.
  - Fail to reject H_0.
  - With a p-value > .05 there is insufficient evidence to suggest COVID impacted Excess Work Hours
- Withhold Documents changes were not statistically significant.
  - Fail to reject H_0.
  - With a p-value > .05 there is insufficient evidence to suggest COVID impacted Withhold Documents.

While most features in Means of Control showed statistical significance, the high raw percent of change is a bit exaggerated compared to the normalized percent of change which were mostly small to moderate shifts. Every feature showed decreased occurrence post-COVID, while Deny Basic Needs may warrant further research.

Relationship to Exploiter

Relationship to Exploiter	Intimate Partner	Friend	Family	Other
% Change Post COVID Raw	-35.66%	-66.48%	14.48%	-47.49%
% Change Post COVID Normalized	-0.98%	-1.46%	0.72%	-3.29%
% Of Total Post COVID	5.55%	1.16%	6.33%	7.43%
% of Total Pre COVID	6.53%	2.62%	5.60%	10.73%
Statistic	45.05489	292.91638	25.05478	343.86906
P-value	1.916E-11	1.1511E-65	5.5724E-07	9.16791E-77
Statistically Significant?	Yes	Yes	Yes	Yes
Degrees of Freedom	1	1	1	1
EF Post COVID-False	44279.264	46218.8226	44368.9145	42768.99409
EF Post COVID-True	2878.7362	939.17743	2789.08545	4389.0059
EF Pre COVID-False	58453.736	61014.1774	5857208545	56460.0059
EF Pre COVID-True	3800.2638	1239.82256	3681.91454	5793.99409

- Intimate Partner changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted Intimate Partner.
  - A decrease of 0.98% in the normalized percent of change represents a small shift post-COVID.
- Friend changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted Friend.
  - A decrease of 1.46% in the normalized percent of change represents a small to moderate shift post-COVID.
- Family changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted Family.
  - An increase of 0.72% in the normalized percent of change represents a small shift post-COVID.
- Other changes were statistically significant.
  - Reject the H_0.
  - With a p-value < .05 there is sufficient evidence to suggest COVID impacted Other.
  - A decrease of 3.29% in the normalized percent of change represents a moderate shift post-COVID.

While all features in Relationship to Exploiter showed statistical significance, the high raw percent of change is a bit exaggerated compared to the normalized percent of change which were mostly small to moderate shifts. Family was the only feature to show an increase, albeit a small shift, which may warrant further research.

To cross-validate the chi-square test results, PCA and k-means were applied to a clean dataset that applied pre-processing methods tailored to the specific needs PCA and k-means require. These methods are detailed in the previous Data and Methods section.

Principal Component Analysis

Limitations encountered in the dataset included features with percentages of missingness > 80%. Due to these limitations, many features were dropped. The only features maintained from the chi-square tests were yearOfRegistration and isSexualExploit. To proceed with meaningful data, features were kept that were not part of the chi-square tests but have direct correlation to type of sexual exploitation (Prostitution) and means of control (Psychological/Physical/Sexual Abuse). To help create a broader picture of changes, two additional features without correlation to chi-square tests were also retained; age and gender.

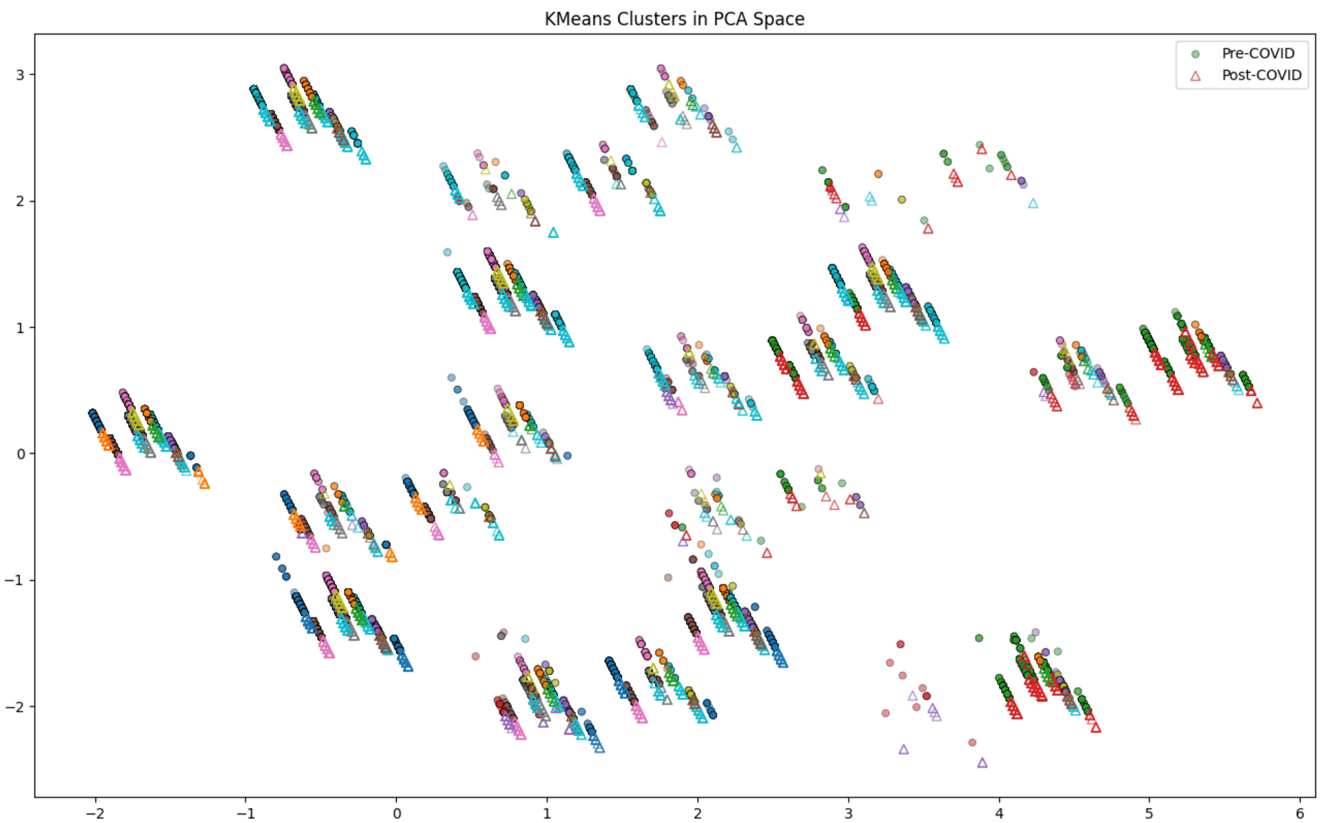
After pre-processing and reducing dimensionality, 15 principal components were evaluated.

PC1	High	Male victims missing sexual exploitation/prostitution fields with some Psychological/Physical/Sexual Abuse indicators present as means of control; slight trend toward recent years.
	Low	Female victims with explicit sexual exploitation and prostitution present and low Psy/Phy/Sex abuse means of control.
PC2	High	Female victims with Psy/Phy/Sex abuse as means of control and prostitution is present; age between 24-26.
	Low	Earlier male victims with missing prostitution and missing Psy/Phy/Sex abuse indicators; age between 9-17.
PC3	High	Female victims between the age of 30-38 with Psy/Phy/Sex abuse as means of control and prostitution indicators are missing; slight trend towards recent years.
	Low	Male and unknown gender victims where age is missing, prostitution indicators are present, and Psy/Phy/Sex abuse means of control are missing.
PC4	High	Recent female victims whose age is unknown, missing prostitution and sexual exploitation indicators, but Psy/Phy/Sex abuse is present as the means of control.
	Low	Victims aged 0-23 with unknown gender and prostitution is present.
PC5	High	Female victims where Sexual Exploitation values are missing, but prostitution is present (data integrity issue as prostitution is a subcategory of sexual exploitation). Psy/Phy/Sex abuse means of control are missing.
	Low	Earlier victims age 0-8 with unknown gender, sexual exploitation is present, and prostitution indicators missing.
PC6	High	Male victims aged 9-17 where sexual exploitation is present, prostitution indicators are missing, and Psy/Phy/Sex abuse is present as the means of control.
	Low	Unknown gender victims, with victims aged 48+, or age is 0-8, or age is 18-20, or age is 21-23; slightly earlier years.
PC7	High	Transgender/NonConforming or male victims aged 18-23 with prostitution indicators missing.
	Low	Earlier cases with victims aged 30-48+.
PC8	High	Victims aged between 30-38 or 18-20, male or transgender/nonconforming, prostitution indicators present; slightly recent years.
	Low	Mostly older ages listed in descending order: 39-47, 48+, 21-23, 24-26, 27-29.
PC9	High	Male victims aged 21-23 or 30-38 with sexual exploitation present.
	Low	Older ages, listed in descending order: 39-47, 48+, 18-20, 27-29; slightly earlier years.
PC10	High	Male victims aged 24-26, 0-8, or 30-38.
	Low	Earlier transgender/nonconforming victims aged 27-29, 21-23, or 39-47.
PC11	High	Transgender/NonConforming victims aged 27-29, or 24-26, or 48+, with Psy/Phy/Sex abuse means of control missing.
	Low	Earlier female victims with a broad range of ages listed in descending order: 39-47, 18-20, 0-8, NaN, 21-23.
PC12	High	Female or unknown gender victims with a broad range of ages listed in descending order: 48+, NaN, 9-17, 30-38.
	Low	Earlier transgender/nonconforming or male victims with a broad range of ages listed in descending order: 39-47, 27-29, 24-26.
PC13	High	Recent transgender/nonconforming victims with sexual exploit indicators present, with a broad range of ages listed in descending order: 39-47, 48+, 21-23, 24-26.
	Low	Female victims with a broad range of ages listed in descending order: 0-8, 27-29, NaN, 18-20.

PC14	High	Recent male victims with sexual exploitation markers present, with a broad range of ages listed in descending order: 18-20, 48+, 24-26, 21-23.
	Low	Younger victims aged 0-8 where gender is either transgender/nonconforming, NaN, female, or age is NaN.
PC15	High	Recent female or transgender/nonconforming victims aged 0-8 or 18-20, with sexual exploitation indicators missing.
	Low	Gender missing, sexual exploitation is present, and ages in descending order are: 30-38, 39-47, 27-29.

K-Means Clustering

K-means unsupervised machine learning methods were then applied to visualize the clusters in the PCA space. While distinct clusters were observed, they were not tightly centralized.



PCA and K-Means Analysis

A heatmap was used to further analyze the properties of the clusters.

The principal components with the strongest post-COVID signals are: PC4, PC13, PC14, PC15 with PC14 and PC15 having the highest post-COVID magnitude.

The principal components with the strongest pre-COVID signals are: PC2, PC5, PC7, PC10, PC11, PC12 with PC7 and PC5 having the highest pre-COVID magnitude.

With the goal of cross-validating the chi-square results, clusters were identified that aligned strongly with the principal components associated with pre-/post-COVID.

Five clusters were identified with this alignment.

- Post-COVID: Cluster 4, Cluster 10
- Pre-COVID: Cluster 2, Cluster 5, Cluster 8

Cluster 4 exhibits strong magnitude in both the highest pre-COVID principal components (PC7 and PC5) and the highest post-COVID principal components (PC14 and PC15), however, the high magnitude in PC13 ultimately pulls this cluster to post-COVID. Further research would be warranted into Cluster 4 as it suggests there may be some patterns that are highly present in both pre- and post-COVID data.

By finding clusters strongly aligned with principal components that support pre- and post-COVID patterns, cross-validation has been provided to support the chi-square test results which also found statistical significance indicating COVID shifted patterns in human trafficking.

Final Observations

With the majority of chi-square tests showing statistical significance between pre- and post-COVID shifts in data, and the cross-validation with principal components and k-means clustering finding natural patterns between pre- and post-COVID data points, the null hypothesis should be rejected.

Observed outcomes showing multiple decreases in features post-COVID were unexpected. To validate the chi-square code and outcomes, further analysis was conducted between two variables outside the scope of the initial chi-square tests, but with outcomes that were expected to be known. Chi-square tests were performed on typeOfSexProstitution and typeOfSexPornography as logically Prostitution would be expected to drop post-COVID and Pornography would be expected to increase. Chi-square tests confirmed statistically significant shifts with prostitution dropping 27% post-COVID and pornography increasing 115% providing confidence in the chi-square results obtained for all other features.

Further chi-square testing highlighted statistically significant shifts in the type of forced labor, with Construction remaining the same, Domestic Workers, and Hospitality decreasing post-COVID and Agriculture increasing 70%.

This analysis provides the validation to support COVID impacting human trafficking in the United States. Further research would be warranted to study the multiple decreases observed across many features. Some research already conducted speaks to the effect COVID had on reduced law enforcement capacity to identify and investigate human trafficking cases (United Nations Office on Drugs and Crime, 2021, p.35). The decreases observed in this analysis may better be explained by a decrease in cases being documented instead of interpreted as a decrease in the actual prevalence of human trafficking occurring post-COVID.

References

Counter Trafficking Data Collaborative. (2025). *The global synthetic dataset*. https://www.ctdatacollaborative.org/page/global-synthetic-dataset

Polaris Project. (2023). *Human trafficking during the COVID and post-COVID era.* https://polarisproject.org/wp-content/uploads/2020/07/Hotline-Trends-Report-2023.pdf

United Nations Office on Drugs and Crime. (2021). *The effects of the COVID-19 pandemic on trafficking in persons and responses to the challenges: a global study of emerging evidence*. United Nations Digital Library.

https://digitallibrary.un.org/record/3931430?v=pdf